Application Settings

Consumer Key (API Key)HTgXiD3kqncGM93bxlBczTfhR
Consumer Secret (API Secret)
djgP2zhAWKbGAgiEd4R6DXujipXRq1aTSdoD9yaHSA8q97G8O
Access LevelRead and write (modify app permissions)
Owner dpnkray
Owner ID 1371497528
and access tokens:
Access Token
1371497582-xD5GxHnkpg8z6k0XqpnJZ3Xvlyc1vVJGUsDXNWZ
Access Token Secret
Qm9tV2XvlOcwbrL2z4QktA3azydtglYPqflZglJ3D4WQ3

Figure 1: Twitter application settings

Here, *requestURL*, *accessURL* and *authURL* are available from the application setting of *https://apps.twitter.com/*.

Connect to Twitter

This exercise requires R to have a few packages for calling all Twitter related functions. Here is an R script to start the Twitter data analysis task. To access the Twitter data through the just created application *myapptwitterR*, one needs to call *twitter, ROAuth* and *modest* packages.

```
>setwd('d:\\r\\twitter')

>install.packages("twitteR")
>install.packages("ROAuth")
>install.packages("modest")

>library("twitteR")
>library("ROAuth")
>library("httr")
```

To test this on the MS Windows platform, load Curl into the current workspace, as follows:

```
>download.file (url="http://curl.haxx.se/ca/cacert.
pem",destfile="cacert.pem")
```

Before the final connectivity to the Twitter application, save all the necessary key values to suitable variables:

```
>consumerKey='HTgXiD3kqncGM93bxlBczTfhR'
>consumerSecret='djgP2zhAWKbGAgiEd4R6DXujipXRq1aTSdoD9yaHSA8
q97G80e'
>requestURL='https://api.twitter.com/oauth/request_token',
>accessURL='https://api.twitter.com/oauth/access_token',
>authURL='https://api.twitter.com/oauth/authorize')
```

With these preparations, one can now create the required

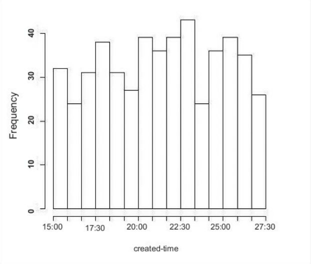

Figure 2: Histogram of created time tag

connectivity object:

```
>cred<- OAuthFactory$new(consumerKey,consumerSecret,requestUR
L,accessURL,authURL)
>cred$handshake(cainfo="cacert.pem")
```

Authentication to a Twitter application is done by the function *setup_twitter_oauth()* with the stored key values as:

```
>setup_twitter_oauth(consumerKey, consumerSecret,AccessToken,
AccessTokenSecret)
```

With all this done successfully, we are ready to access Twitter data. As an example of data analysis, let us consider the simple problem of opinion mining.

Data analysis

To demonstrate how data analysis is done, let's get some data from Twitter. The Twitter package provides the function *searchTwitter()* to retrieve a tweet based on the keywords searched for. Twitter organises tweets using hashtags. With the help of a hashtag, you can expose your message to an audience interested in only some specific subject. If the hashtag is a popular keyword related to your business, it can act to increase your brand's awareness levels. The use of popular hashtags helps one to get noticed. Analysis of hashtag appearances in tweets or Instagram can reveal different trends of what the people are thinking about the hashtag keyword. So this can be a good starting point to decide your business strategy.

To demonstrate hashtag analysis using R, here, we have picked up the number one hashtag keyword *#love* for the study. Other than this search keyword, the *searchTwitter()* function also requires the maximum number of tweets that the function call will return from the tweets. For this discussion, let us consider the maximum number as 500. Depending upon